Programs on Problem Solving

a: Solve the 8 Queens Problem

Aim

To place 8 queens on an 8x8 chessboard so that no two queens attack each other, i.e., no two queens share the same row, column, or diagonal.

Explanation:

This is a classic **Constraint Satisfaction Problem (CSP)**. We solve it using **backtracking**, which means placing queens row by row, checking safety, and backtracking when a conflict arises.

Procedure:

1. Create an 8x8 board initialized with 0s.
2. Place a queen in each row by checking if it's safe.
3. If no column is safe in a row, backtrack.
4. Repeat until all 8 queens are placed.

Python Code:

```
N = 8

def print_solution(board):
    for row in board:
        print(" ".join("Q" if col else "." for col in row))
    print("\n")

def is_safe(board, row, col):
    for i in range(row):
        if board[i][col] or \
            (col - (row - i) >= 0 and board[i][col - (row - i)]) or \
            (col + (row - i) < N and board[i][col + (row - i)]):
            return False
    return True

def solve_n_queens(board, row):
    if row == N:
        print_solution(board)
        return True
    found = False
    for col in range(N):
        if is_safe(board, row, col):
            board[row][col] = 1
            found = solve_n_queens(board, row + 1) or found
            board[row][col] = 0
    return found
```

```
board = [[0] * N for _ in range(N)]
solve_n_queens(board, 0)
```

Output :

```
. . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .


Q . . . . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
. . . . . Q . .
. . . Q . . . .
. Q . . . . . .
. . . . Q . . .


Q . . . . . . .
. . . . . . Q .
. . . Q . . . .
. . . . . Q . .
. . . . . . . Q
. Q . . . . . .
. . . . Q . . .
. . Q . . . . .


Q . . . . . . .
. . . . . . Q .
. . . . Q . . .
. . . . . . . Q
. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . Q . . . . .


. Q . . . . . .
. . . Q . . . .
. . . . . Q . .
. . . . . . . Q
. . Q . . . . .
Q . . . . . . .
. . . . . . Q .
. . . . Q . . .


. . . . . . . Q
. . Q . . . . .
Q . . . . . . .
. . . . . Q . .
. Q . . . . . .
. . . . Q . . .
```

```
. . . . . . Q .
. . . Q . . . .


. . . . . . . Q
. . . Q . . . .
Q . . . . . . .
. . Q . . . . .
. . . . . Q . .
. Q . . . . . .
. . . . . . Q .
. . . . Q . . .


True
```

Result:

Thus the program of placing 8 queens on an 8x8 chessboard was executed successfully.